ExamLens

Index -

Topic	Page No.
Objective	2
Details of the Project (Opportunity, problems, solution, Market)	2
Pre-requisites	8
Complete System Workflow	9
Visual WorkFlows	11
All Paid API Requirements	17
Requirements	19
Explanations	20
Link Of the Current Build	21

Objective –

ExamLens is a behavioural diagnostic intelligence platform for JEE aspirants. The core objective is to answer the question every student has after a mock test but no platform currently answers — not what went wrong, but why it went wrong.

The platform records every interaction a student makes during a test (time per question, answer changes, skips, revisits, which wrong option was chosen) and applies a two-layer analysis engine. Layer 1 uses rule-based scoring to classify the error type from behaviour alone. Layer 2 uses LLM-generated question metadata to identify the exact misconception, wrong belief, and specific fix. The platform additionally runs a spaced repetition revision scheduler that maintains a live mastery map across all 80+ JEE topics per student and adjusts daily revision plans automatically.

The seed fund is required to move from the current working prototype (built and deployed at edu-insights-hub.vercel.app) to a production-ready system capable of handling the first paid coaching institute pilot.

Details of the Project (Opportunity, problems, solution, Market) -

The Opportunity

India produces more than 2.5 million JEE aspirants every year.

Over the last decade, almost every part of the preparation ecosystem has been digitized. Students have access to unlimited lectures, notes, doubt-solving platforms, previous-year papers, mock tests, and study material from anywhere in the country.

Content is no longer scarce.

A student in a small town has access to nearly the same learning resources as a student studying in Kota.

Yet despite taking dozens of mock tests, many students plateau for months.

Their scores fluctuate, but the underlying mistakes remain unchanged.

They continue losing marks to the same misconceptions, the same calculation errors, the same revision gaps, and the same test-taking behaviours.

The issue is not a lack of content.

The issue is that students are expected to diagnose themselves.

The Problem

Every test platform today tells students what happened.

Students see:

- Marks
- Rank
- Percentile
- Chapter-wise accuracy
- Subject-wise analysis
- Time spent

But these numbers do not explain why the score happened.

A student scoring 142/300 could be losing marks because:

- they lack conceptual understanding,
- repeatedly make calculation errors despite knowing the concept,
- change correct answers under pressure,
- misread questions,
- apply the wrong formula despite knowing the topic,
- spend too much time on low-value questions,
- leave solvable questions unattempted,
- repeatedly fall for the same misconception,
- or are simply forgetting previously learned material due to poor revision timing.

Each of these problems requires a completely different intervention.

Current platforms treat them identically because they evaluate only the final answer.

They can't tell students what happened.

They do not tell them what to do next.

As a result, students leave most test reviews with information, but not direction.

Why Existing Analysis Falls Short

Today's testing platforms are optimized for evaluation, not diagnosis.

They answer questions such as:

- What score did the student get?
- What percentile did they achieve?

But they do not answer the questions that actually improve performance:

• Why did marks drop?
• Which mistakes are recurring?
• Which mistakes are costing the most marks over time?
• Which weaknesses require study and which require behavioural correction?
• What specific action should the student take next?

Two students can score the same marks for completely different reasons.

Existing platforms treat them identically.

ExamLens does not.

Why This Market Gap Exists

The obvious question is:

“If this is valuable, why doesn’t it already exist?”

The answer lies in how the Indian test-preparation market evolved.

Even today, a large percentage of coaching institutes—including many of the most respected brands—still conduct tests using printed papers and OMR sheets.

An OMR sheet records only one thing:

The final answer.

It captures none of the behavioural data required to understand how a student arrived at that answer.

There is no record of:

• time spent per question,
• answer changes,
• skipped questions,
• revisits,
• section-switching behaviour,
• or decision-making under pressure.

Without behavioural data, diagnostic intelligence is impossible.

The industry spent years optimizing evaluation.

Almost nobody built the intelligence layer on top of evaluation.

That is the gap ExamLens is addressing.

Why Now

The transition from paper-based testing to computer-based testing is creating an entirely new category of data.

For the first time, student decision-making inside a test can be captured and analyzed at scale.

Every click, answer change, revisit, skip, and timing decision becomes measurable.

This creates an opportunity that did not exist a decade ago.

And now, every student can use phone, they can use the app for the tests, and see their analysis, while faculty and see on their desk.

ExamLens is built to transform this behavioural data into actionable intelligence.

We Are Not Another EdTech Platform

We're not competing with Allen, Aakash, PW, Unacademy, or coaching institutes.

We're building a layer that can sit underneath any of them.

Whether a student studies from YouTube, Kota modules, school teachers, coaching institutes, or online learning platforms is irrelevant to us.

As long as they take tests, we can generate value.

The content ecosystem is already mature.

We are focused on understanding performance.

Our goal is not to replace teachers or content providers.

Our goal is to become the diagnostic intelligence layer for competitive-exam preparation.

Our Solution

ExamLens is a diagnostic intelligence platform for competitive-exam preparation.

We record every interaction a student makes during a test:

- time spent per question,
- answer changes,
- skipped questions,
- revisits,
- section navigation,

• selected distractors,
• and test-stage behaviour.

From these signals, we identify recurring patterns behind performance.

Instead of saying:

“You lost 18 marks in Physics.”

We can say:

“You have lost 74 marks across your last six tests because you repeatedly change correct answers during the final 20 minutes.”

Or:

“You consistently select the same misconception-linked distractor in Electrostatics questions.”

The focus shifts from score reporting to root-cause analysis.

The system not only identifies problems but prioritizes them based on their actual impact on performance.

The Revision Layer

The second problem is retention.

A typical JEE student must manage more than 80 topics over a two-year preparation cycle.

Students know they should revise.

What they do not know is:

• what to revise,
• when to revise,
• what is at risk of being forgotten,
• and which topics will deliver the highest return on revision time.

ExamLens maintains a live mastery map of the syllabus for every student.

Using performance history and memory-decay models, the platform automatically:

• schedules revision,
• adjusts for missed days,
• reprioritizes weak topics,
• identifies topics at risk of being forgotten,
• and shifts strategy as the exam approaches.

The student does not need to manually analyze tests or build revision schedules.

The platform tells them exactly what to do next.

Value for Coaching Institutes

Our primary customer is not the individual student.

It is the coaching institute.

Every institute already conducts tests.

We are not asking them to change their pedagogy.

We are adding a diagnostic layer on top of an activity they already perform.

For faculty, the biggest challenge is not conducting tests.

It is identifying what to do after the test.

A batch of 300 students can generate thousands of incorrect responses.

Reviewing them manually is impossible.

ExamLens converts that data into action.

Instead of asking faculty to inspect hundreds of answer sheets, the platform automatically surfaces:

- the concepts causing the largest mark losses,
- misconceptions shared across the batch,
- topics that require reteaching,
- students who need academic intervention,
- students whose issue is behavioural rather than conceptual,
- and the concepts most likely to improve outcomes if addressed immediately.

The goal is not better reporting.

The goal is helping faculty make better teaching decisions in less time.

Business Model

We operate on a per-student subscription model through coaching institutes.

Example:

• 300 students
• ₹x per student per month

Revenue:

Approx. ₹12,000 per month from a single institute.

For institutes charging ₹1.5–3 lakh per student annually, this cost is negligible relative to the value provided.

More importantly, each institute serves three purposes simultaneously:

1. Revenue source
2. Distribution channel
3. Reference customer

A successful deployment creates proof of value, making expansion into additional institutes significantly easier.

Pre-Requisites –

What already exists and is working:

The platform is live and functional. The test engine records behavioural events. Layer 1 rule-based tag scoring is implemented in TypeScript with all 9 error tags, gates, and conflict resolution. Layer 2 LLM question metadata JSON is generated and integrated. Post-test review screen shows diagnosis per question. Action Board detects cross-test patterns. Syllabus engine with SM-2 spaced repetition is running. Faculty admin panel allows question paper upload. AI Mentor chat is functional. Parent report generation works. A 30-question test paper with full distractor metadata JSON has been validated and is live.

What is not yet done and requires the fund:

Question metadata JSON for a full JEE question bank (requires Mathpix API + LLM API costs for bulk processing). Production-grade infrastructure capable of handling 200–2000 concurrent students. Mathpix OCR integration for PDF paper parsing at scale. Supabase production tier upgrade for the database. Vercel and Render production tier for hosting. Domain, SSL, and basic marketing. Mobile-responsive UI polish.

Complete System Workflow -

Workflow A — Faculty uploads a question paper

Faculty logs into the admin panel → uploads a PDF or Word document → Python backend receives the file → for PDF: Mathpix API extracts text and identifies diagram/equation regions as image crops → for Word doc: python-docx parser extracts text and inline images → questions are chunked into batches of 10 → each batch is sent to LLM API (Gemini/Claude) with the structured prompt → LLM returns a JSON array with chapter, topic, subtopic, difficulty, distractor analysis per option (distractor_type, misconception, wrong_belief, reasoning_path, confidence), is_calculation_heavy, trap_keyword flag → JSON is validated programmatically → all 30/75/90 questions stored in Supabase with metadata → faculty reviews flagged low-confidence questions → test goes live.

Workflow B — Student takes a test

Student opens test → every interaction is logged to Supabase in real time (question_enter, answer_selected, answer_changed with timestamp, question_flagged, question_exit, question_revisit, section_switch, test_submit) → timer runs client-side → answer sequence with timestamps stored per question → test submitted → behavioural event stream finalised.

Workflow C — Post-test analysis (Layer 1)

System reads behavioural events for each wrong answer → computes: time_vs_avg_ratio, answer_change_count, revisit_count, was_flagged, had_correct_answer_at_any_point, time_held_correct_answer → reads question metadata JSON from database → pre-step: if distractor_type null or confidence < 0.65, return fallback Conceptual Gap LOW → else run all 9 tag scoring blocks → conflict resolution: if top two scores within 8 points, flag for Adaptive Clarification → highest-scoring tag assigned → stored with source = "rule_based".

Workflow D — Post-test analysis (Layer 2)

For each tagged wrong answer, system reads the stored distractor JSON for the specific option the student chose → surfaces: misconception, wrong_belief, reasoning_path, recommended_revision → rewrites to second-person conversational tone before display → shown alongside Layer 1 tag in post-test review screen → student taps Yes/No to verify → corrections stored with source = "student_corrected" and 1.5x weight for future ML training.

Workflow E — Adaptive Clarification Engine

Triggered when Layer 1 tag confidence gap <= 8 points between top two tags → system identifies top two candidate error types → constructs LLM prompt with: question text, correct answer, student's chosen distractor, top 2 candidate tags, student's mastery and behavioural data → LLM generates 10 targeted micro-questions designed to discriminate between the two candidate errors → student completes the set → responses fed back to

scoring engine → confident diagnosis produced → stored with source = "clarification_engine".

Workflow F — Mastery update

After each test, for every question answered: direction = +1 (correct) or -1 (incorrect) → difficulty_weight applied (Easy wrong = 1.3x, Medium = 1.0x, Hard wrong = 0.7x, inverses for correct) → base_step = 8 → mastery_delta = direction × 8 × difficulty_weight → new_mastery = clamp(old_mastery + delta, 0, 100) → cannot exceed 90 until 5+ real test-derived data points exist → topic state updated in Supabase.

Workflow G — Revision scheduling

SM-2 algorithm runs per student per topic → ease factor and inter-repetition interval updated after each Hard/Good/Easy rating → forgetting curve R(t) = e^(-t/S) used to identify topics at risk (R < 0.70) → daily revision queue generated → missed-day recovery redistributes overdue topics across next available days → exam countdown mode shifts strategy at <30 days → revision-lock at <10 days → "studied" toggle updates topic state without affecting mastery score.

Workflow H — Cross-test pattern detection (Action Board)

After every test completion, system aggregates all error tags across student's full test history → identifies same error_type on same topic appearing in 2 tests (Watching state) or 3+ tests (Confirmed state) → for confirmed patterns: calculates cumulative marks lost estimate, generates plain-language "What is Happening" description, generates "Exact Next Step" with specific resource and exercise count → sorted by cumulative marks lost descending → displayed on Action Board.

Workflow I — Faculty batch report

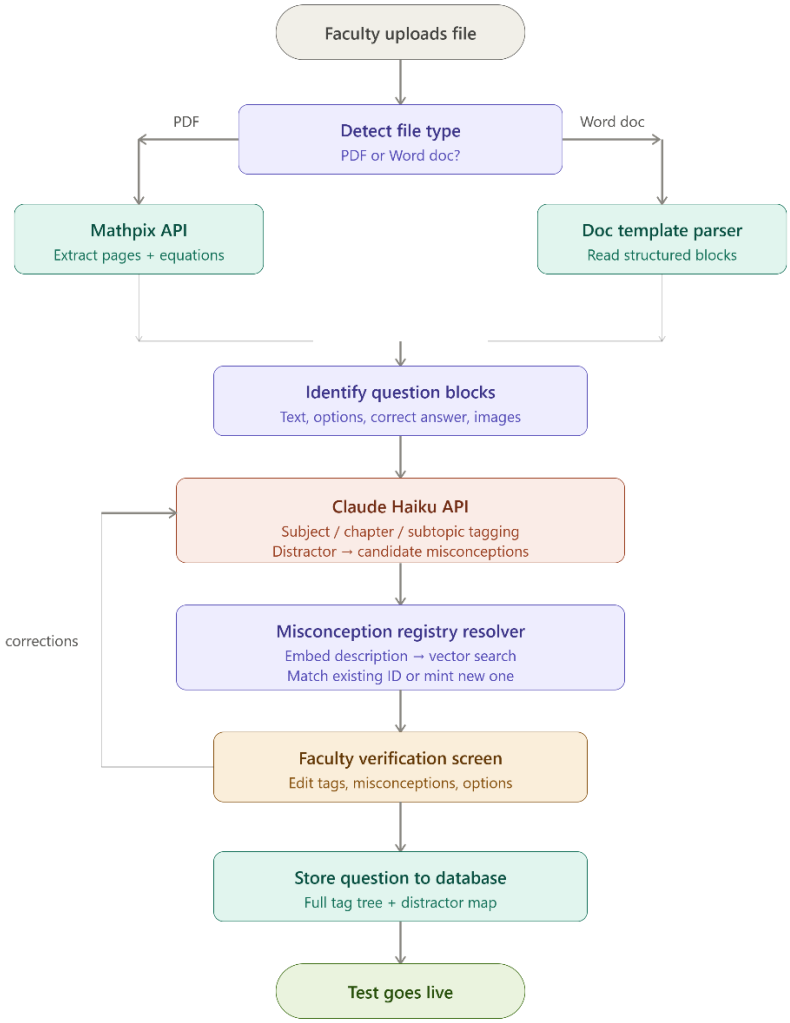
After multiple students complete the same test, system aggregates: topic-wise wrong rate (flags topics where >40% of batch wrong) → distractor frequency heatmap (most chosen wrong options across batch mapped to shared misconceptions) → student cluster segmentation (K-means on performance vectors into Performing / Borderline / Needs Attention) → Reteach alerts generated automatically → individual student drill-down available.

Workflow J — Parent report

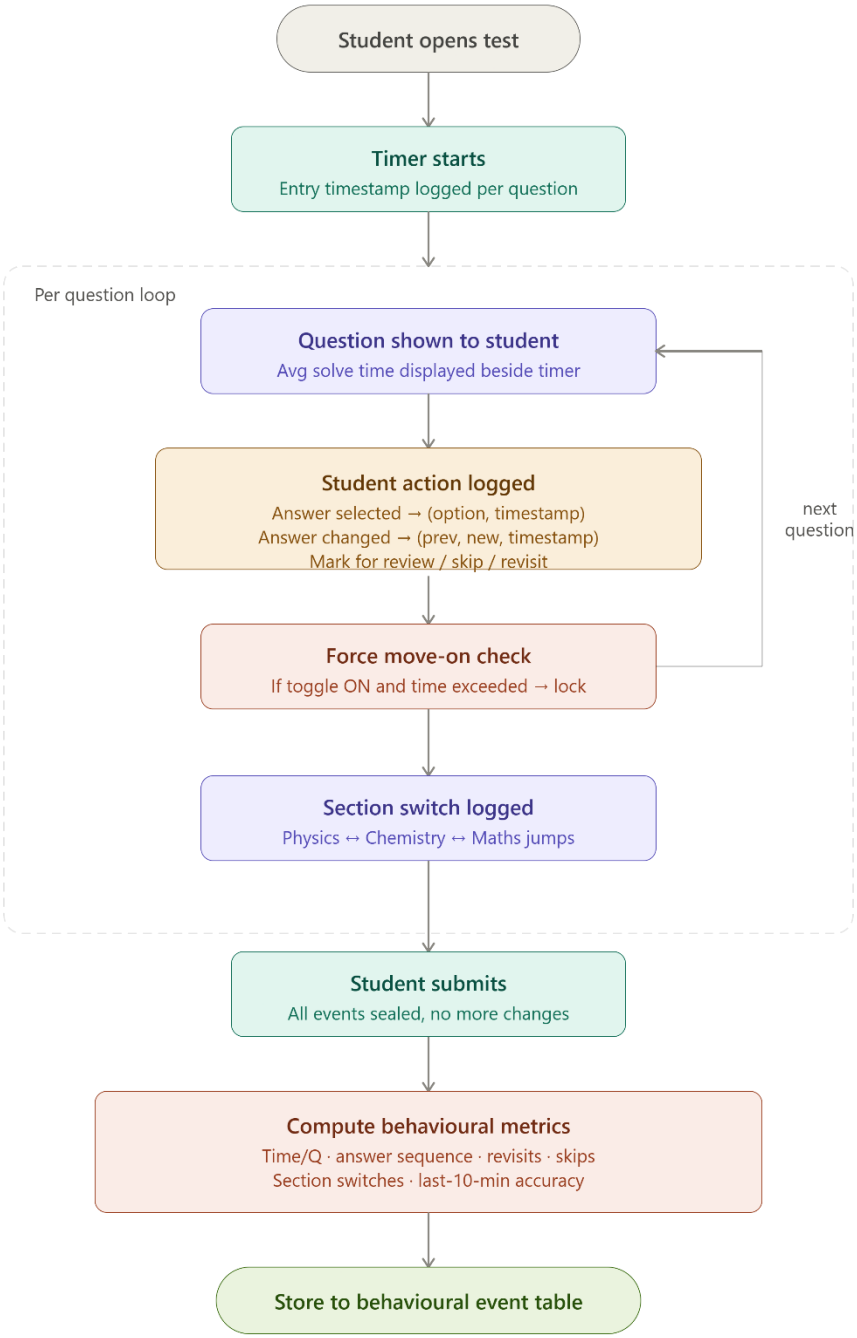
Weekly: system computes effort score from login frequency, tests attempted, revision queue completion → LLM generates plain-language summary from student data → trajectory (improving/stagnant/declining) computed from score trend → delivered via email or in-app.

Visual WorkFlows –

Question paper upload — file type detection, Mathpix/parser branch, Haiku tagging, misconception registry resolver, faculty verification loop

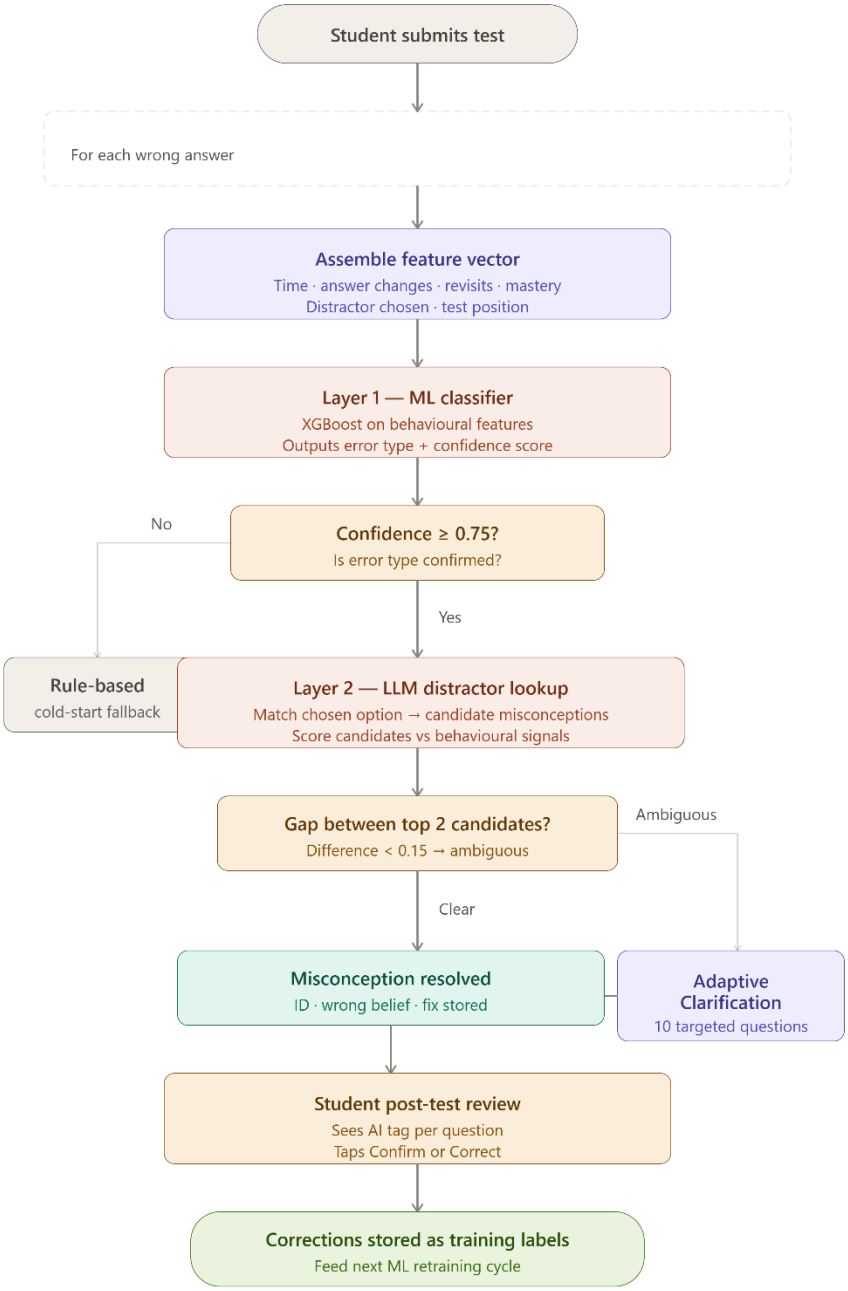


Student takes test — per-question event logging, force move-on check, section switches, submission and metric computation

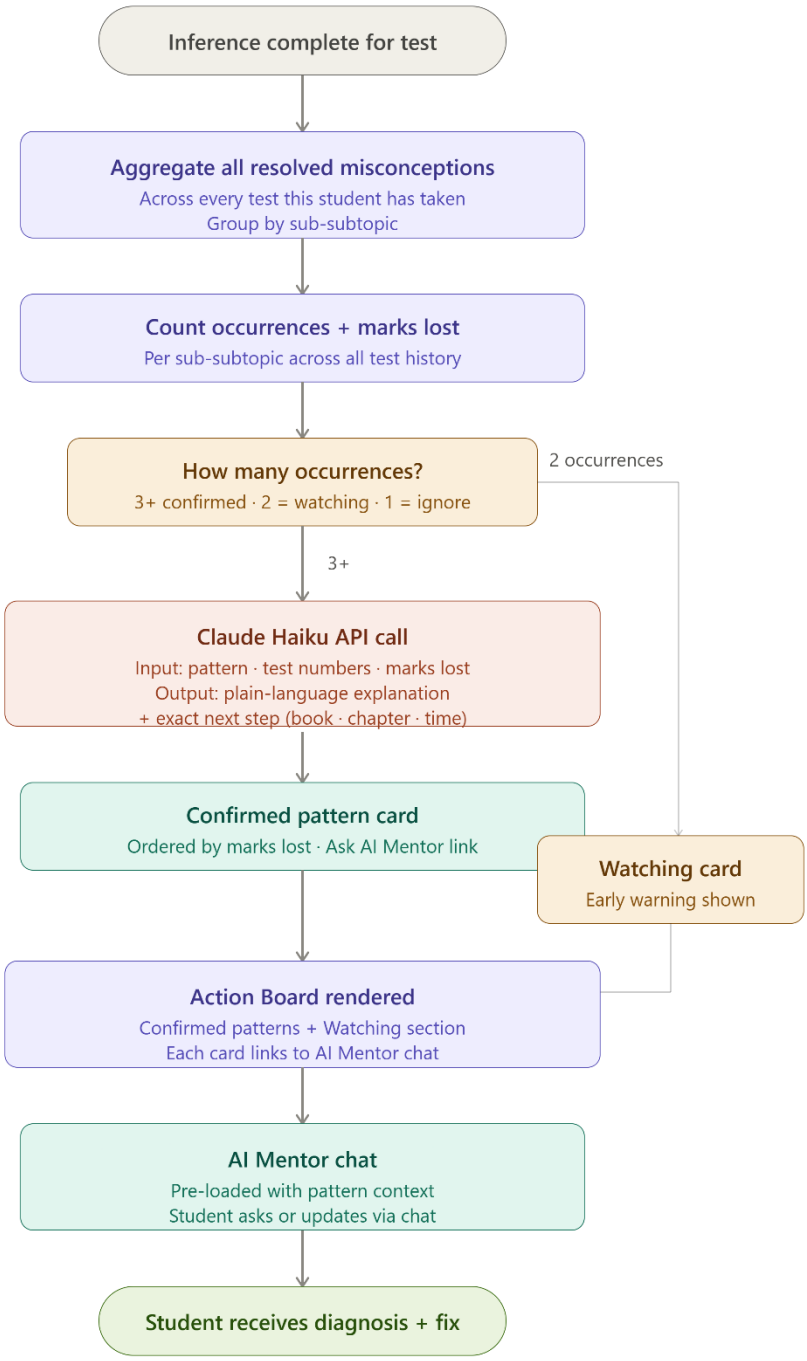


Post-test inference — feature vector assembly, Layer 1 ML classifier, confidence check, Layer 2 LLM distractor lookup,

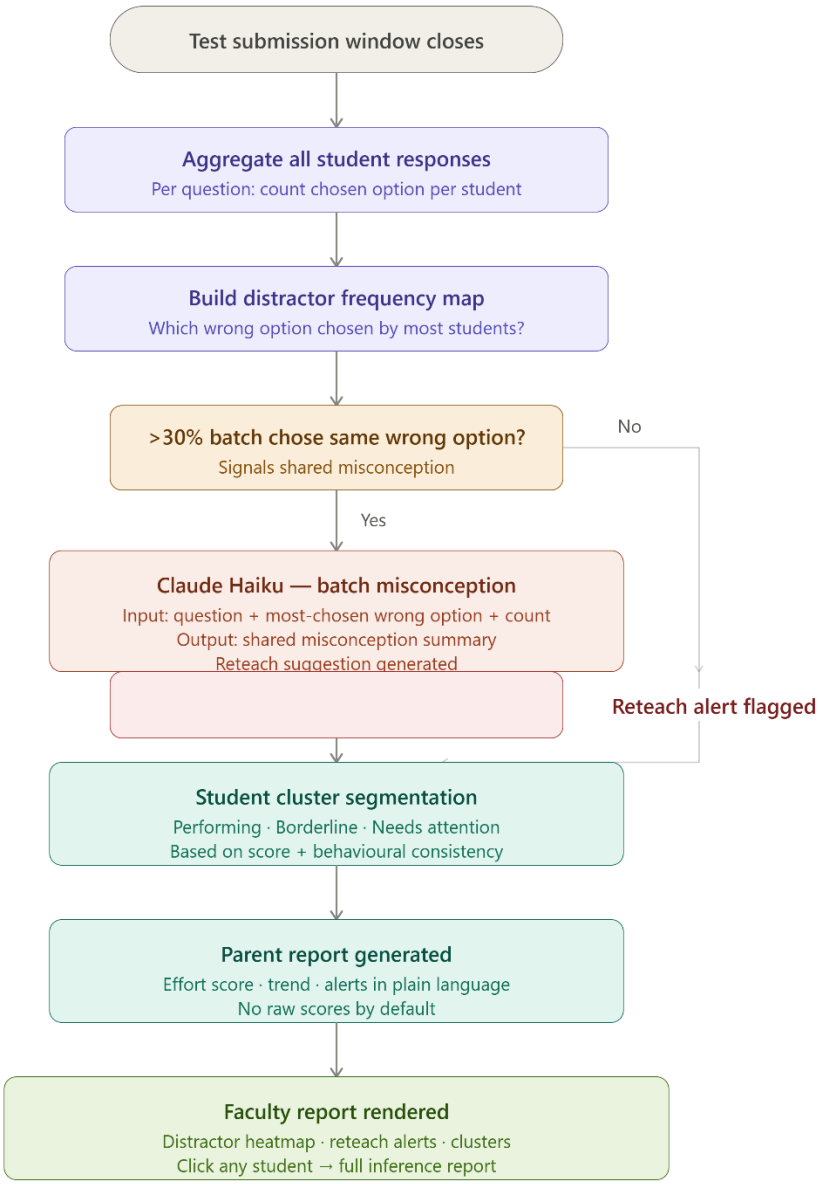
ambiguity → Adaptive Clarification Engine branch, student verification as training data



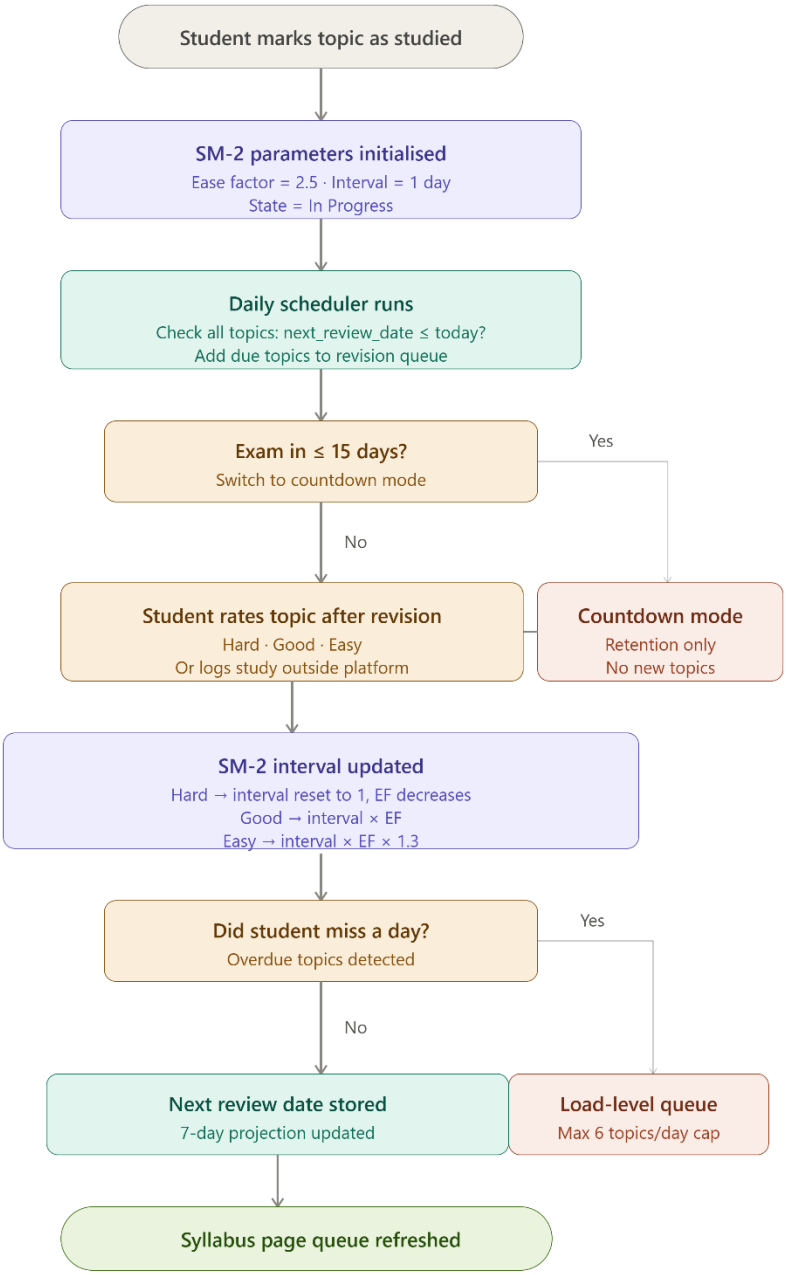
Action Board generation — cross-test aggregation, occurrence thresholding, Haiku plain-language call, confirmed vs watching cards, mentor chat link



Faculty batch report — distractor frequency mapping, 30% threshold for reteach alert, LLM batch misconception call, student cluster segmentation, parent report



Spaced revision scheduler — SM-2 initialisation, daily scheduler, exam countdown mode at ≤15 days, Hard/Good/Easy rating → interval update, missed day load-levelling


Student marks topic as studied
SM-2 parameters initialised
Ease factor = 2.5 · Interval = 1 day
State = In Progress
Daily scheduler runs
Check all topics: next_review_date ≤ today?
Add due topics to revision queue
Exam in ≤ 15 days?
Switch to countdown mode
Yes
No
Student rates topic after revision
Hard · Good · Easy
Or logs study outside platform
Countdown mode
Retention only
No new topics
SM-2 interval updated
Hard → interval reset to 1, EF decreases
Good → interval × EF
Easy → interval × EF × 1.3
Did student miss a day?
Overdue topics detected
Yes
No
Next review date stored
7-day projection updated
Load-level queue
Max 6 topics/day cap
Syllabus page queue refreshed

All Paid API Requirements -

API 1 — Mathpix API

Purpose: OCR for question paper PDFs. Mathpix handles mathematical notation, equations, diagrams, and handwritten content that standard OCR cannot parse. Required for the PDF upload pipeline (PYQ papers, Allen/Aakash test series PDFs).
When called: Once per PDF at upload time. Not called during tests or analysis.
Cost structure: $0.004 per page (approximately ₹0.33 per page). A 90-question JEE paper is typically 20–30 pages.

Cost per paper: ₹7–10 per paper upload.

Monthly estimate (bare minimum): Processing 10 papers per month = ₹100/month.
Monthly estimate (institute scale, 200 students, 4 tests/month): 4 papers × ₹10 = ₹40/month. Mathpix costs are negligible at this scale.
Annual bare minimum: ₹1,200

API 2 — LLM API (Gemini Flash / Claude)

Purpose: Two uses. First: generating question metadata JSON at upload time (distractor analysis, chapter tagging, difficulty rating) — called once per question, result cached permanently. Second: Adaptive Clarification Engine micro-question generation — called when tag confidence gap <= 8 points.

When called for metadata: Once per paper upload, in batches of 10 questions per API call.
Cost per 10-question batch: Approximately ₹2–3 (input ~2000 tokens, output ~1500 tokens at Gemini Flash pricing).

Key point: Once a question's metadata is generated and cached, it is never re-generated. If 500 students attempt the same JEE Main 2024 paper, the API is called exactly once for that paper, not 500 times.

Current status: Gemini free tier is being used. Must migrate to paid for production (rate limits, no SLA, data privacy concerns as flagged earlier).

API 3 — Supabase (Database)

Purpose: PostgreSQL database hosting all student data, test data, behavioural events, question metadata, mastery scores, revision schedules.

Current tier: Free (500MB storage, limited connections).
Why paid tier is required: Free tier has a 500MB database limit and connection pooling limits that will be exceeded by 50+ concurrent students. The behavioural event table alone generates approximately 200 rows per student per test (every click logged). 200 students × 10 tests × 200 events = 400,000 rows in events table alone, before questions, answers, and metadata.
Paid tier (Pro): $25/month (approximately ₹2,100/month). Includes 8GB storage, connection pooling, daily backups, no pausing.
Annual cost: ₹25,200

API 4 — Vercel (Frontend Hosting)

Purpose: Hosts the React frontend.
Current tier: Free (hobby). Serverless function execution limits and no SLA on free tier.
Why paid matters: Free tier has 100GB bandwidth limit and no team collaboration. At 200 students each loading the app 10 times per day, bandwidth consumption is moderate but the lack of SLA is a risk during exam periods.
Pro tier: $20/month (approximately ₹1,680/month).
Honest assessment: Vercel free tier may be sufficient through the pilot phase. Upgrade only when hitting limits.
Annual cost if upgraded: ₹20,160. Can defer this cost.

API 5 — Render (Backend Hosting)

Purpose: Hosts the FastAPI Python backend (ML engine, Gemini calls, document parsing, parent report generation).
Current tier: Free tier (spins down after inactivity — 15-second cold start on first request).
Why paid is needed: Cold starts are unacceptable during live tests. If a student submits a test and the backend is spun down, the first analysis request takes 15 seconds to respond. Students will think the platform crashed.
Starter tier: $7/month (approximately ₹590/month). Always-on, no cold starts.
Annual cost: ₹7,080

API 6 — Anthropic Claude API

Purpose: Gemini Flash quality on distractor analysis is insufficient, Claude is the alternative. Also used for the AI Mentor chat in the current codebase.

Cost: Similar to Gemini Flash. Approximately ₹2–4 per 10-question batch.
When to use: If Gemini distractor confidence scores are consistently below 0.65 (causing excessive fallbacks), switch to Claude for metadata generation.

Requirements -

First Coaching Institute (200–2000 Students, ₹20,000/month Revenue Target)

This assumes one or two coaching institutes are onboarded, with 200 to 2000 active students, each taking 2–4 tests per month.

Item	Purpose	Monthly (200 students)	Monthly (2000 students)
Render (Standard tier)	Higher memory, faster processing for concurrent analysis	₹2,500	₹5,000
Supabase Pro	8GB storage, connection pooling	₹2,500	₹2,500
Supabase compute upgrade	Larger instance for concurrent queries	₹0	₹4,200
LLM API — Gemini/Claude	Metadata (new papers) + clarification engine at scale	₹1,500	₹8,000
Mathpix API	10–20 new papers/month from institutes	₹500	₹1000
Domain	.com/.org	₹2000/year	₹2000/year
Vercel Pro / own hosting	Bandwidth + team collaboration + SLA	₹2000	₹3000
Email delivery (Resend/SendGrid)	Parent weekly reports, student notifications	₹500	₹2,000
Error monitoring (Sentry free tier)	Catch production bugs before students report them	₹0	₹0
Antigravity Pro	Fast AI Coding	₹1950	₹1950

Claude Code	Ai Coding And metadata generattion	₹2000	₹2000
Total infrastructure	-	**₹11,630/month**	**₹27,730/month**

Note – Domain price is not Included in this total.

Explanation of Critical Items -

Why Supabase Pro is non-negotiable: The free tier pauses the database after 1 week of inactivity and has a 500MB limit. A paused database during a student's test is catastrophic. Pro tier guarantees uptime and has enough storage for approximately 2 years of full student data at 200-student scale.

Why Render paid is non-negotiable: The Python backend handles all LLM calls, document parsing, and ML inference. On the free tier it spins down after 15 minutes of inactivity. The cold start delay appears to students as the platform crashing immediately after test submission. This single issue would destroy trust in the first demo.

Why Mathpix and not free OCR: Free OCR (Tesseract, Google Vision) fails on mathematical notation — fractions, integrals, vector notation, chemical equations. A question paper where "$\int x^2 dx$" is OCR'd as "fx2dx" produces unusable distractor analysis. Mathpix is purpose-built for academic mathematics and is the only viable option for JEE question papers.

Why LLM metadata is a one-time cost per question: Once generated and cached, the distractor analysis for a question is never regenerated. Processing JEE Main 2025 (90 questions) costs ₹22.50 once. Every student who ever attempts that paper uses the cached metadata. As the question bank grows, the marginal cost of adding a new paper decreases.

Why the Gemini free tier must be replaced: The free tier has a daily limit of approximately 1,500 requests. At 200 students each triggering analysis on 20 wrong answers, with 10% requiring clarification engine calls (400 calls), the daily limit is hit within one busy test day. Beyond rate limits, the free tier has no data privacy guarantees — student test performance data sent through a free consumer API is not acceptable for a commercial product used by coaching institutes.

Link Of The Current Build –

https://edu-insights-hub.vercel.app/